Jorge Cardenas-Gamboa, M.Sc.

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Summary

- Computational physicist with a focus on Materials Design. Expert in high-performance computing (HPC) and large-scale DFT simulations. Proven track record in modeling light-matter interactions and electronic structures to accelerate the development of next-generation technologies.

Employment History

2024 – Present
- **PhD Researcher**, Leibniz Institute for Solid State and Materials Research (IFW Dresden), Germany - Optimized numerical models for light-matter interactions in chiral crystals, leading to a new method for enantiomer selectivity.

2022 – 2024
- **PhD Researcher**, Max Planck Institute for Chemical Physics of Solids (MPI-CPfS), Germany - Executed and managed large-scale DFT simulations on HPC clusters, implementing Python scripts to automate data extraction and enhance processing efficiency. Collaborated with experimental groups to guide material synthesis, using predictive simulations tools.

Education

2022 – Present
- **PhD in Theoretical and Computational Physics** TU Dresden, Germany.

2020 – 2022
- **MSc in Theoretical Chemistry and Computational Modelling (Erasmus Mundus)** Universitat de Barcelona & Sorbonne Université.

2014 – 2020
- **BSc in Chemistry** Yachay Tech University, Ecuador.

Skills

Coding
- Python (Numpy, Pandas, Scipy, Matplotlib & others), Bash, Fortran, Git, LaTeX, Jupyter Notebook.

Numerical Software
- VASP, Quantum ESPRESSO, FPLO, WannierTools, Wannier90, VESTA.

Languages
- English (Professional), German (A1 – actively learning), Spanish (Native).

Professional
- Research, problem-solving, scientific writing, collaboration, teaching.

Awards and Achievements

2025
- **Visiting Researcher**, Université de Sherbrooke, Canada

2023 & 2024
- **Visiting Researcher**, DIPC Institute

2020
- **Scholarship**, Erasmus Mundus Fellowship for Master in Theoretical Chemistry and Computational Modelling

2021
- **Summer School**, DIPC Institute

2018
- **Internship**, Laboratoire de Physique et Chimie Théoriques, Université de Lorraine

Research Publications

Journal Articles

1. **J. Cardenas-Gamboa**, E. Lesne, A. Ernst, M. G. Vergniory, P. McClarty, and C. Felser, “Exploring non-collinear magnetic ground states in tetragonal Mn_2-based heusler compounds,” *Journal of Physics: Materials*, vol. 9, no. 1, p. 015 005, 2026.
2. H. Lin et al., “Decoupling structural and electronic dimensionality: 2D transport in a 3D honeycomb chiral stacking,” *Matter*, 2026.
3. J. Wang et al., “Quantifying quasiparticle chirality in a chiral topological semimetal,” *arXiv preprint arXiv:2603.14640*, 2026.
4. P. Yanda et al., “Direct evidence of topological dirac fermions in a low carrier density correlated 5d oxide,” *Advanced Functional Materials*, vol. 36, no. 13, e12899, 2026.
5. **J. Cardenas-Gamboa** et al., “Photostriction-driven phase transition in layered chiral $NbOX_2$ crystals: Electrical-field-controlled enantiomer selectivity,” *arXiv preprint arXiv:2510.12998*, 2025.
6. A. Chimarro-Contreras, Y. Lopez-Revelo, **J. Cardenas-Gamboa**, and T. Terencio, “Insights into the effect of charges on hydrogen bonds,” *International Journal of Molecular Sciences*, vol. 25, no. 3, p. 1613, 2024.
7. A. M. Ochs et al., “KMg_4Bi_3: A narrow band gap semiconductor with a channel structure,” *Inorganic Chemistry*, vol. 63, no. 43, pp. 20 133–20 140, 2024.